1.5.1 Trend (depends on time linearly) and seasonality (P.6) (1)

$$\mu_t = E(X_t)$$

Trend: Long term change in μ_t (e.g. $\mu_t = \alpha + \beta t$)

Seasonality: Regular periodic variation in μ_t

(e.g. $\mu_t = \mu_{t+4}$ or $\mu_t = \mu_{t+12}$)

seasonal — monthly

$$X_t = \underbrace{\mu + \alpha t}_{\text{trend}} + \underbrace{\beta \cos \omega t + \gamma \sin \omega t}_{\text{seasonal}} + \underbrace{Z_t}_{\text{error}}$$

where $Z_t \sim N(0, \sigma^2)$ $\quad iid \equiv$ independent and identically distributed

1.5.2 Stochastic trend (the time element is statistically variate)

Eg. $X_t = X_{t-1} + Z_t$ where $Z_t \sim NID(0, \sigma^2)$

normally and independently distributed

If we let $X_0 = 0$, then

$$X_1 = X_0 + Z_1 = 0 + Z_1 = Z_1$$

$$X_2 = X_1 + Z_2 = Z_1 + Z_2$$

$$X_3 = X_2 + Z_3 = Z_1 + Z_2 + Z_3 = \sum_{n=1}^{3} Z_n$$

$$\vdots$$

$$X_t = \sum_{n=1}^{t} Z_n \sim N(0, t\sigma^2)$$

$$E(X_t) = E\left(\sum_{n=1}^{t} Z_n\right) = \sum_{n=1}^{t} E(Z_n) = \sum_{n=1}^{t} 0 = 0.$$

$$V(X_t) = V\left(\sum_{n=1}^{t} Z_n\right) = \sum_{n=1}^{t} V(Z_n) = \sum_{n=1}^{t} \sigma^2 = t\sigma^2$$

Differencing operation (p.6 bottom)

$$\nabla X_t = X_t - X_{t-1} \quad \text{from } X_t = X_{t-1} + Z_t$$
$$= Z_t \sim N(0, \sigma^2) \qquad X_t - X_{t-1} = Z_t \text{ (p.7 top)}$$

It can be seen that differencing can eliminate stochastic trend (from our example).

Similarly, if

$$X_t = \mu + \alpha t + Z_t \quad \text{— (1) where } Z_t \sim NID(0, \sigma^2)$$

(equation (1))

$$X_{t-1} = \mu + \alpha(t-1) + Z_{t-1}$$
$$= \mu + \alpha t - \alpha + Z_{t-1} \quad \text{— (2)}$$

—)

$$\nabla X_t = X_t - X_{t-1} = \alpha + Z_t - Z_{t-1}$$

(1) − (2) gives

$$E(\nabla X_t) = E(\alpha + Z_t - Z_{t-1}) = \alpha$$

$$V(\nabla X_t) = V(\alpha + Z_t - Z_{t-1}) = V(Z_t - Z_{t-1})$$
$$= V(Z_t) + V(-Z_{t-1}) = V(Z_t) + (-1)^2 V(Z_{t-1})$$
$$= \sigma^2 + \sigma^2 = 2\sigma^2$$

i.e. $\nabla X_t \sim NID(\alpha, 2\sigma^2)$

Similarly, for seasonal differencing, for our example (equation (1) above)

$$\nabla_k X_t = X_t - X_{t-k} \sim NID(k\alpha, 2\sigma^2)$$
$$= X + \alpha t + Z_t - [\mu + \alpha(t-k) + Z_{t-k}] = k\alpha + Z_t - Z_{t-k}$$

1.5.3 Causal model and regression over time (p.7)

(A) Linear model

$$Y_t = \alpha + \beta X_t + \varepsilon_t \qquad \text{Causal model}$$

$$Y_t = \alpha + \beta t + \varepsilon_t \qquad \text{Trend model}$$

ε_t satisfies (i) $\varepsilon_t \sim N(0, \sigma^2)$

(ii) ε_t's are independent

To test (ii), the presence of autocorrelations

H_0: ε_t's are independent

H_1: ε_t's are correlated (not independent)

Durbin-Watson test statistic

$$d = \frac{\sum_{t=2}^{n} (e_t - e_{t-1})^2}{\sum_{t=1}^{n} e_t^2}$$

$$= \frac{\sum_{t=2}^{n} (e_t^2 - 2e_t e_{t-1} + e_{t-1}^2)}{\sum_{t=1}^{n} e_t^2}$$

$$\cong \frac{2\sum_{t=2}^{n} e_t^2 - 2\sum_{t=2}^{n} e_t e_{t-1}}{\sum_{t=1}^{n} e_t^2}$$

N.B. $\dfrac{\sum_{t=2}^{n} e_t^2}{\sum_{t=1}^{n} e_t^2} \cong 1$

$$\cong 2\left(\frac{\sum_{t=2}^{n} e_t^2}{\sum_{t=1}^{n} e_t^2} - \frac{\sum_{t=2}^{n} e_t e_{t-1}}{\sum_{t=1}^{n} e_t^2} \right)$$

roughly

$$\cong 2(1 - \hat{r}_1)$$

$\hat{r}_1$ is the correlation coefficient between e_t and e_{t-1}.

Note that $0 < d < 4$, with a value close to 0 indicates a +ve correlation and a value close to 4 indicates a -ve correlation. (4)

Two more values need to calculate to complete the testing procedure. They are d_L and d_U.

$H_0 : \rho = 0$

$H_1 : \rho > 0$ or $\rho < 0$

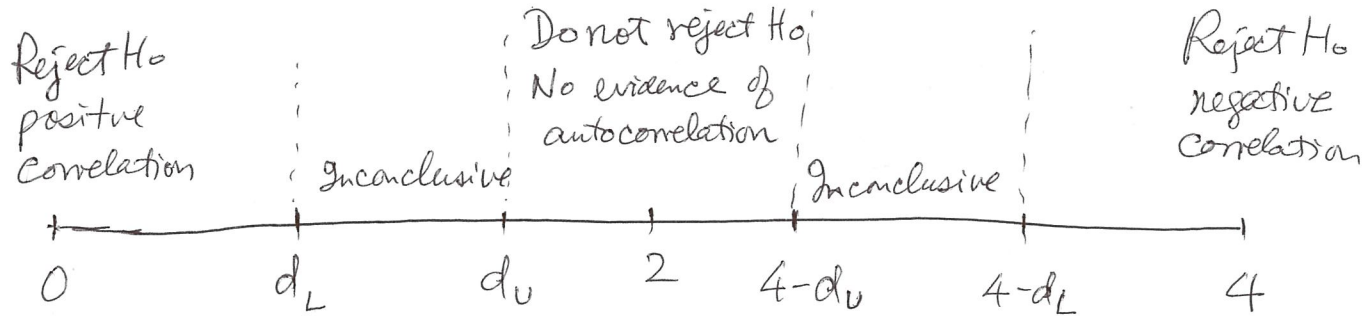
(1) If $d_U < d < 4 - d_U$, do not reject H_0 (i.e, accept H_0) (pg top)

(2) If $d < d_L$, reject H_0 (i.e., accept $\rho > 0$)

If $d > 4 - d_L$, reject H_0 (i.e., accept $\rho < 0$)

(3) If $d_L < d < d_U$ or $4 - d_U < d < 4 - d_L$,

then the test is inconclusive (i.e., no conclusion)



(for p.8) (5)

Ordinary Least Squares Method (OLS)

We will find a line of 'best fit', we will minimize the Sum of Squares Errors (SSE):

$$Y_i = \alpha + \beta X_i + \varepsilon_i \qquad \Rightarrow \qquad \varepsilon_i = Yi - \alpha - \beta X_i$$

$$SSE = \sum_{i=1}^{n} \varepsilon_i^2 = \sum_{i=1}^{n} (Y_i - \alpha - \beta X_i)^2 .$$

This is minimized with respect to α and β when

$$\frac{\partial SSE}{\partial \alpha} = \sum_{i=1}^{n} (Y_i - \hat{\alpha} - \hat{\beta} X_i)(1) = 0 \qquad \Rightarrow \qquad \sum_{i=1}^{n} Y_i = n\hat{\alpha} + \hat{\beta} \sum_{i=1}^{n} X_i \qquad \text{(a)}$$

$$\frac{\partial SSE}{\partial \beta} = \sum_{i=1}^{n} (Y_i - \hat{\alpha} - \hat{\beta} X_i)(X_i) = 0 \qquad \Rightarrow \qquad \sum_{i=1}^{n} X_i Y_i = \hat{\alpha} \sum_{i=1}^{n} X_i + \hat{\beta} \sum_{i=1}^{n} X_i^2 \qquad \text{(b)}$$

Dividing (a) by n, we get $\overline{Y} = \hat{\alpha} + \hat{\beta}\overline{X}$.

Multiplying (a) by $\dfrac{\sum_{i=1}^{n} X_i}{n}$ and subtracting from (b),

$$\sum_{i=1}^{n} X_i Y_i = \hat{\alpha} \sum_{i=1}^{n} X_i + \hat{\beta} \sum_{i=1}^{n} X_i^2$$

$$-) \qquad \frac{\left(\sum_{i=1}^{n} X_i\right)\left(\sum_{i=1}^{n} Y_i\right)}{n} = \hat{\alpha} \sum_{i=1}^{n} X_i + \hat{\beta} \frac{\left(\sum_{i=1}^{n} X_i\right)^2}{n}$$

$$\sum_{i=1}^{n} X_i Y_i - \frac{\left(\sum_{i=1}^{n} X_i\right)\left(\sum_{i=1}^{n} Y_i\right)}{n} = \hat{\beta}\left[\sum_{i=1}^{n} X_i^2 - \frac{\left(\sum_{i=1}^{n} X_i\right)^2}{n}\right]$$

Therefore an estimator of β, the slope of the regression line, is

$$\hat{\beta} = \frac{\sum_{i=1}^{n} X_i Y_i - \dfrac{\left(\sum_{i=1}^{n} X_i\right)\left(\sum_{i=1}^{n} Y_i\right)}{n}}{\sum_{i=1}^{n} X_i^2 - \dfrac{\left(\sum_{i=1}^{n} X_i\right)^2}{n}} = \frac{\sum_{i=1}^{n} (X_i - \overline{X})(Y_i - \overline{Y})}{\sum_{i=1}^{n} (X_i - \overline{X})^2} = \frac{S_{xy}}{S_{xx}} \left[\text{or } \frac{C_{xy}}{C_{xx}} \right]$$

$$= \frac{\sum_{i=1}^{n} (X_i - \overline{X}) Y_i}{\sum_{i=1}^{n} (X_i - \overline{X})^2}$$

and an estimator of α, the y-intercept, is $\hat{\alpha} = \overline{Y} - \hat{\beta}\overline{X}$.
Hence the best-fit line (the regression of Y on X) is

$$\hat{Y}_i = \hat{\alpha} + \hat{\beta}X_i, \; i = 1, 2, \ldots, n.$$

$\frac{S_e^2}{C_{xx}}$

Both $\hat{\alpha}$ and $\hat{\beta}$ are unbiased estimators.

$$V(\hat{\beta}) = \frac{\sigma^2}{\sum_{i=1}^{n}(X_i - \bar{X})^2} \qquad V(\hat{\beta}) = \frac{\hat{\sigma}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \hat{\sigma}_{\hat{\beta}}^2 = S_{\hat{\beta}}^2 = S_b^2$$

$$= \frac{\sigma^2}{S_{xx}} = \frac{\sigma^2}{C_{xx}} \quad \text{where } \hat{\sigma}^2 = \frac{\sum (Y_i - \hat{\alpha} - \hat{\beta}x_i)^2}{n-2} = \frac{\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2}{n-2}$$

$$= \frac{\sum_{i=1}^{n} e_i^2}{n-2} = \frac{SSE}{n-2} = MSE$$

↓ mean square error

Assume σ^2 unknown

$H_0: \beta = 0$

$H_1: \beta \neq 0$ or $\beta > 0$ or $\beta < 0$

$$t_{n-2} = \frac{\hat{\beta} - \beta}{\hat{\sigma}_{\hat{\beta}}} = \frac{\hat{\beta} - \beta}{S_b} \overset{\text{under } H_0}{=\!=} \frac{\hat{\beta}}{S_b}$$

depends on time t

Causal Model (in terms of X)

For the trend model, we have (in our notes)

$$Y_t = \alpha + \beta t + \varepsilon_t$$

[Note that X is replaced by t in the causal model]

$$\hat{\beta} = b = \frac{C_{yt}}{C_{tt}} = \frac{\sum (y_t - \bar{y})(t - \bar{t})}{\sum (t - \bar{t})^2} = \frac{\sum (y_t - \bar{y})t}{\sum (t - \bar{t})^2}$$

$$\hat{\alpha} = a = \bar{y} - b\bar{t}$$

For the test of the slope, $H_0: \beta = 0$ vs $H_1: \beta \neq 0$, or $\beta > 0$ or $\beta < 0$

$$t_{n-2} = \frac{b - \beta}{S_b}, \text{ where } S_b = \sqrt{\frac{MSE}{C_{tt}}} = \sqrt{\frac{\frac{\sum e_t^2}{n-2}}{\sum (t - \bar{t})^2}}$$

Forecasting with linear trend

Model : $Y_{t+\tau} = \alpha + \beta(t+\tau) + \varepsilon_t$

The OLS estimate of $Y_{t+\tau}$ is

$$\hat{Y}_{t+\tau} = \hat{\alpha} + \hat{\beta}(t+\tau) = a + b(t+\tau)$$

where $\sum \varepsilon_t^2$ is minimized.

The following is to find the variance of $\hat{Y}$ before we want to find the variance of $\hat{Y} - Y_0$ for the prediction interval. $\hat{Y}_t = \hat{\alpha} + \hat{\beta} x_t$

For $\hat{Y}|X=x_0 = \hat{\alpha} + \hat{\beta} x_0$, and it can be shown that

$E(\hat{Y}|X=x_0) = \alpha + \beta x_0$ is unbiased and with variance

$$V(\hat{Y}|X=x_0) = \sigma^2\left(\frac{1}{n} + \frac{(x_0-\bar{X})^2}{\sum_{i=1}^{n}(x_i-\bar{X})^2}\right)$$

$$E(\hat{Y}|X=x_0) = E(\hat{\alpha}+\hat{\beta}x_0) = E(\hat{\alpha}) + E(\hat{\beta})x_0 = \alpha + \beta x_0$$

since $\hat{\alpha}$ and $\hat{\beta}$ are unbiased estimators of α and β respectively.

$$V(\hat{Y}|X=x_0) = V(\hat{\alpha}+\hat{\beta}x_0) = V(\bar{Y} - \hat{\beta}\bar{X} + \hat{\beta}x_0)$$

$$= V[\bar{Y} + \hat{\beta}(x_0-\bar{X})] = V(\bar{Y}) + (x_0-\bar{X})^2 V(\hat{\beta})$$

$$= \frac{\sigma^2}{n} + (x_0-\bar{X})^2 \frac{\sigma^2}{\sum_{i=1}^{n}(x_i-\bar{X})^2} = \sigma^2\left[\frac{1}{n} + \frac{(x_0-\bar{X})^2}{\sum_{i=1}^{n}(x_i-\bar{X})^2}\right]$$

$\therefore \widehat{V(\hat{Y}|X=x_0)} = \hat{\sigma}^2\left[\frac{1}{n} + \frac{(x_0-\bar{X})^2}{\sum_{i=1}^{n}(x_i-\bar{X})^2}\right]$ (this is an unbiased estimator of $V(\hat{Y}|X=x_0)$)

estimate of $V(\hat{Y}|X=x_0)$

In time series analysis, we used to forecast future values, hence prediction for Y is useful. (p. 8)

When $X = x_0$, $Y = Y_0$, where $Y_0 = \alpha + \beta x_0 + \varepsilon_0$. Thus we have

$$\hat{Y}_0 = \hat{\alpha} + \hat{\beta} x_0 = \bar{Y} + \hat{\beta}(x_0 - \bar{X})$$
$$\bar{Y} - \hat{\beta}\bar{X} \qquad = \bar{Y} + \hat{\beta} x_0 - \hat{\beta}\bar{X}$$

To find a prediction interval for Y_0, we consider the random variable $\hat{Y}_0 - Y_0$ and have

$$E(\hat{Y}_0 - Y_0) = E\{(\hat{\alpha} + \hat{\beta} x_0) - (\alpha + \beta x_0 + \varepsilon_0)\}$$
$$= E(\hat{\alpha}) + E(\hat{\beta}) x_0 - \alpha - \beta x_0 - E(\varepsilon_0)$$
$$= \alpha + \beta x_0 - \alpha - \beta x_0 - 0 = 0$$

$\therefore$, $\hat{Y}_0$ is unbiased for Y_0.

$$V(\hat{Y}_0 - Y_0) = V\{[\bar{Y} + \hat{\beta}(x_0 - \bar{X})] - (\alpha + \beta x_0 + \varepsilon_0)\}$$
$$= V(\bar{Y}) + (x_0 - \bar{X})^2 V(\hat{\beta}) + V(-\varepsilon_0) \qquad (-1)^2 V(\varepsilon_0) = +1 \cdot \sigma^2 = \sigma^2$$
$$= \frac{\sigma^2}{n} + \frac{(x_0 - \bar{X})^2 \sigma^2}{\sum_{i=1}^{n} (x_i - \bar{X})^2} + \sigma^2$$
$$= \sigma^2 \left[1 + \frac{1}{n} + \frac{(x_0 - \bar{X})^2}{\sum_{i=1}^{n} (x_i - \bar{X})^2} \right]$$

Causal model

$\therefore$, $\widehat{V(\hat{Y}_0 - Y_0)} = \hat{\sigma}^2 \left[1 + \frac{1}{n} + \frac{(x_0 - \bar{X})^2}{\sum_{i=1}^{n} (x_i - \bar{X})^2} \right]$

For our time series prediction interval, we replace all the X's by t's, thus

Trend model

$$\widehat{V(y_t)} = S_e^2 \left[1 + \frac{1}{n} + \frac{(t_p - \bar{t})^2}{\sum_{i=1}^{n} (t_i - \bar{t})^2} \right], \quad C_{tt} = \sum_{i=1}^{n} (t_i - \bar{t})^2 \text{ in our notes.}$$

$\frac{\sum e_t^2}{n-2} = \frac{SSE}{n-2}$ ↗ Note that t_p is given

(B) Curvilinear model

Tools for detection of nonlinearity:

1. Look at the original Time Series plot.
2. Make use of Durbin-Watson's test for residuals from linear regression

$$d \cong 2(1-\hat{r}_1), \qquad \hat{r}_1 = \frac{\sum_{t=2}^{n} e_t e_{t-1}}{\sum_{t=1}^{n} e_t^2}$$

where $\hat{r}_1$ is the correlation coefficient.

3. Plot of 1st differences, i.e., $X_t - X_{t-1}$

If the original data possess only linear trend, then the 1st difference will roughly give you a random graph.

If the original data come from quadratic function, then observe

$$X_t = a + bt + ct^2$$

$$X_{t+1} = a + b(t+1) + c(t+1)^2$$

$$X_{t+1} - X_t = a + b(t+1) + c(t+1)^2 - a - bt - ct^2$$

$$= b + 2ct + c$$ it is linear trend

which implies the existence of trend.
Thus the presence of trend in the 1st difference indicates a nonlinear model.

Other Model types: (p.9)

(I) Polynomial:

$$y_t = a + b_1 t + b_2 t^2 + \cdots + b_k t^k + \varepsilon_t,$$

$$t = 1, 2, \ldots, n.$$

This is a multiple regression with

$$Y_{n\times 1} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \qquad \beta_{(k+1)\times 1} = \begin{pmatrix} a \\ b_1 \\ b_2 \\ \vdots \\ b_k \end{pmatrix} \qquad \varepsilon_{n\times 1} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

$$X_{n\times(k+1)} = \begin{pmatrix} 1 & 1 & 1^2 & \cdots & 1^k \\ 1 & 2 & 2^2 & \cdots & 2^k \\ 1 & 3 & 3^2 & \cdots & 3^k \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & t & t^2 & \cdots & t^k \end{pmatrix}$$

$$Y = X\beta + \varepsilon \Rightarrow \varepsilon = Y - X\beta$$

with $E(\varepsilon) = 0_{n\times 1}$ and $V(\varepsilon) = \sigma^2 I_{n\times 1}$

We are going to minimize $S = \sum_{i=1}^{n} \varepsilon_i^2$ with respect to β. In matrix notation

$$S = \varepsilon^T\varepsilon = (Y - X\beta)^T(Y - X\beta)$$

$$= Y^TY - \beta^TX^TY - Y^TX\beta + \beta^TX^TX\beta$$

Please see Lemma 2.2

$$= Y^TY - 2\beta^TX^TY + \beta^TX^TX\beta$$

$$\hookrightarrow \frac{dS}{d\beta} = 0 - 2X^TY + 2X^TX\beta \overset{\text{set}}{=} 0 \Rightarrow X^TY = X^TX\beta$$

$\therefore$ $X^TX\beta = X^TY$ Multiply both sides by $(X^TX)^{-1}$ on the left hand side

$$\hat{\beta} = (X^TX)^{-1}X^TY$$

(II) Exponential:

Log linear models:

(i) $y_t = e^{a+bt}\varepsilon_t$, where $\varepsilon_t \sim N(1, \sigma^2)$

$$\ln y_t = a + bt + \ln \varepsilon_t$$

$$= a + bt + \eta_t, \text{ where } \eta_t = \ln \varepsilon_t \sim N(0, \sigma_1^2)$$

η eta

$$\widehat{\ln y_t} = \hat{a} + \hat{b}t$$

(ii) $y_t = ab^t\varepsilon_t$

$$\ln y_t = \ln a + t\ln b + \ln \varepsilon_t = \ln a + t\ln b + \eta_t$$

$$\widehat{\ln y_t} = \widehat{\ln a} + t\widehat{\ln b} \qquad \eta_t \sim N(0, \sigma_1^2)$$

(iii) $y_t = a t^b \varepsilon_t$

$$\ln y_t = \ln a + b \ln t + \ln \varepsilon_t$$

$$\widehat{\ln y_t} = \widehat{\ln a} + \hat{b} \ln t + \cancel{\eta_t}^{(0)}, \qquad \eta_t \sim N(0, \sigma_1^2)$$

(II) Logistic models

$$y_t = \frac{L}{1 + \exp(\alpha + \beta t + \varepsilon_t)}$$

$$\Rightarrow \hat{y}_t = \frac{L}{1 + \exp(\hat{\alpha} + \hat{\beta} t)} \qquad \varepsilon_t \sim N(0, \sigma^2)$$